

Unpacking Empathy Development in HCI Learners: Patterns from Diary Reflections and Peer Discussions

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Abstract

Understanding user experience and user needs is an essential goal in HCI education, yet novice students often struggle to recognize the diversity and complexity of user perspectives. This study investigates how empathy-related understanding unfolds during early HCI learning through diary reflections and peer discussions. Ten undergraduate students recorded their daily experiences using a fitness application and later shared their reflections in small-group discussions. We analyzed open-ended responses together with multi-stage Empathy in Design Scale data. Our findings center on three aspects. The four empathy dimensions, Emotional Interest, Sensitivity, Personal Experience, and Self-awareness, showed descriptive shifts rather than a single trend. Students gradually realized that their initial assumptions about user needs and design difficulty were not sufficient. They also moved from a function-oriented view toward a more holistic understanding of product experience. These findings offer early insight into how reflective writing and peer dialogue influence the development of empathy.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); • Applied computing → Education.

Keywords

Diary Study, Focus Group Discussion, Empathy

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1 Introduction

Supporting students in understanding user experience and user needs has long been an important objective in human-computer interaction education [18], yet it remains challenging for novices. Novice students often lack the experience and methodological grounding needed to bridge the gap between abstract concepts and real-world contexts. Olesen et al. have shown that computing students encounter many learning difficulties in HCI design education, including confusion about design concepts and uncertainty about how to use design resources effectively [25]. Empathy in engineering and computing education is commonly described as a multidimensional construct that integrates emotional engagement, perspective taking, and reflective distancing, supporting learners in reconsidering assumptions about users and design contexts [1, 22]. Developing empathy has been described as one promising way to support this understanding, as it involves taking another person's standpoint and recognizing their emotions, thoughts, and perspectives [20]. However, although empathy plays an important role in understanding user experience, relatively little is known about how empathy-related understanding unfolds during early HCI learning.

Diary studies represent a practical way to support reflective engagement with technology use. Participants document their interactions with selected technologies together with their accompanying reflections [15, 29], offering insight into how thoughts and behaviors evolve over time. Prior work suggests that lived experience activities, such as diary studies, can help students reflect on user perspectives [16, 32]. The Plan-Collect-Analyze-Reflect (PCAR) diary study framework highlights the importance of structured reflection within diary-based learning [9]. In addition to individual reflection, peer-based activities also offer value. Focus group discussions provide opportunities for students to compare perspectives and consider diverse viewpoints [6, 17]. Combining diary writing with group discussion may therefore create a setting where students engage both individually and socially with questions about user experience [10].

To explore how empathy-related understanding develops in this dual reflective setting, we invited ten undergraduate computer science students to participate in a two-week diary study followed

by focus group discussions. Students documented their daily experiences using a fitness application and later shared and compared their reflections in small group discussions. We collected open-ended responses together with multi-stage Empathy in Design Scale data [7]. Our analysis focuses on identifying patterns across the four empathy dimensions and examining how students described their developing understanding in their reflections. Our findings highlight three emerging patterns in how novice HCI learners develop empathy-related understanding: students recalibrate assumptions about user needs and design difficulty, empathy-related dimensions evolve unevenly across reflective stages, and learners shift from a function-centered perspective toward a more holistic interpretation of product experience. This work offers early insight into how diary-based reflection and peer dialogue may shape emerging empathy in novice HCI learners.

2 Background

2.1 Diary Studies

Diary studies are a qualitative method used in HCI to capture users' experiences and reflections over time [5, 24]. Participants record their interactions with technologies, including contexts, actions, and emotions, either through open-ended writing or structured prompts [9]. These first-person accounts reveal longitudinal changes and subtle aspects of everyday experience that short-term or lab-based methods may overlook [26].

Diary studies have also gained attention in HCI education because they encourage learners to reflect on their own technology use and notice contextual factors that shape user experience [11]. Fan et al. proposed the PCAR (Plan, Collect, Analyze, Reflect) framework to guide the use of diary studies in education [9]. Within this four-stage structure, the Reflect stage plays an important role in helping students revisit and consolidate their diary content to consider user needs and potential human factors, while also prompting them to recognize the limits of their own perspectives rather than treating their personal experiences as a stand-in for users'. Recent work has also explored tools that support reflection, such as timeline-based review systems [3], automated contextual capture [19], and model-based diary summarization [23]. These developments have expanded the role of diary studies as a reflective learning approach in HCI education.

2.2 Focus Group Discussions

Focus group discussions (FGDs) are a qualitative method used in HCI to explore participants' perspectives through group interaction [14]. Unlike individual interviews, they create a social context in which participants respond to one another, compare viewpoints, and surface ideas or concerns that may not arise in one-on-one settings [28]. Although group dynamics may introduce risks such as groupthink [4], FGDs are often employed alongside other methods to provide richer and more reliable insights. In HCI research, they have been used to evaluate technologies, identify user needs, and examine the social dimensions of technology use [27, 30].

FGDs have also been used to support reflective and collaborative learning [13, 21]. Through conversation, students articulate their own ideas, hear alternative interpretations, and reconsider their positions. Prior work shows that such peer interaction can

encourage engagement, critical thinking, and awareness of multiple viewpoints [2, 12]. In our study, FGDs follow the diary activity, enabling students to revisit their individual reflections while encountering diverse experiences from peers. This two-step process complements individual sensemaking with collective perspective taking and provides a richer basis for exploring how students reason about user experience in HCI learning.

3 Method

We selected Keep, a widely used fitness app in China, to offer diverse interaction experiences, including AI-generated training plans, real-time feedback, motivational features, and community engagement. Such variety offered multiple entry points for observing user experience and user needs.

Ten undergraduate computer science students (8 male, 2 female) participated in the study. Participants were novice HCI learners and had not taken any formal coursework in human-computer interaction or user experience design prior to the study. Before starting, we introduced the study goals and demonstrated the application. Students then used a structured diary template that prompted them to describe their daily interactions with the app, note positive and problematic features, evaluate motivational and AI-enabled functions, and reflect on social interactions. Each student completed ten diary entries across two weeks.

After the diary phase, two researchers independently reviewed the entries and conducted preliminary coding. This process surfaced both expected topics (such as reactions to AI feedback and motivational mechanisms) and unexpected ones (such as hardware software compatibility issues or ethically uncomfortable social interactions). A codebook was created, and representative excerpts were selected for group discussion. Students were randomly assigned to two groups with different moderators. To preserve individual viewpoints, each participant first wrote brief responses before the discussion. Each FGD lasted about one hour, during which moderators balanced speaking opportunities, asked follow-up questions, and took notes on key remarks.

We used the Empathy in Design Scale (EIDS) [7] to observe descriptive patterns across four empathy dimensions: Emotional Interest and Discovery (curiosity about users), Sensitivity and Immersion (open minded involvement in users' situations), Personal Experience and Connection (drawing on one's own experiences), and Self awareness and Detachment (distinguishing personal assumptions from users' perspectives). To complement the scale, open-ended questions invited students to reflect on perceived underestimation or overestimation of empathy, benefits of diary writing and discussion, and suggestions for improvement. Data were collected at three points: before the study (EIDS only), after the diary phase (EIDS), and after the FGDs (EIDS plus open-ended responses).

4 Findings

Students' reflections and EIDS responses together revealed notable shifts in their understanding of user experience and user needs throughout the activity. Our analysis identified three main themes: (1) recalibration of under- and overestimation, (2) empathy-related shifts across multiple dimensions, and (3) movement from a

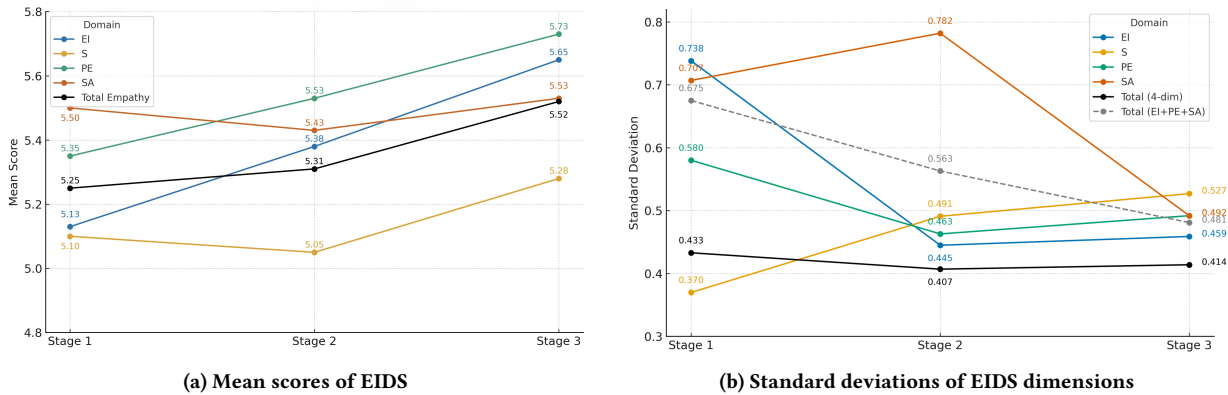


Figure 1: Empathy in Design Scale Results

function-centered view toward a more holistic understanding of product experience.

4.1 Recalibration of Underestimation and Overestimation

The first trend is the recalibration of underestimation and overestimation. All participating students noted that, before beginning the diary study, they underestimated the difficulty of understanding user needs and overestimated the clarity of their own design judgments. As the diary study progressed, they became increasingly aware that the complexity of user experience exceeded their initial expectations and required attention to detail across different contexts. One student reflected:

“At the beginning, I thought understanding user needs was not difficult, but writing the diaries made me realize that many details are actually easy to overlook.”

One student noted how the discussion broadened their view:

“Through discussion, I saw that others faced very different problems with the same app, which made me realize design situations are more complex than I had thought.”

These reflections indicate that both diary writing and group dialogue prompted students to reconsider their assumptions about user needs and to recognize the limits of relying solely on personal experience. Together, the two activities encouraged a more layered view of user experience without assuming that any single account provides a complete picture.

4.2 Empathy-related Shifts Across Multiple Dimensions

The second trend concerns shifts in students’ empathy-related understanding, reflected in both their open-ended reflections and the descriptive patterns of the EIDS responses. Students described gaining a clearer sense of how others might think and experience technology as they moved through the diary and discussion activities. Writing diaries encouraged them to reflect on their own reactions and to imagine challenges or needs that different users

might face. Group discussions added an additional layer by revealing how peers approached similar situations differently, helping students recognize that users cannot be treated as a single, uniform type. One student noted:

“The group discussion helped me revisit my diary insights. On sensitive topics like AI, I saw that users with different habits held different attitudes, which corrected my one-sided views.”

One student noted the activities’ complementary roles:

“There are many ways to learn about user experiences, but none are as impactful as direct experience. Writing diaries helped me step into the user’s perspective, and the discussions made me curious about why others’ experiences differed from mine.”

These qualitative insights align with descriptive patterns observed across the four EIDS dimensions (Figure 1a). EI and PE showed upward trends across the three stages, suggesting increasing curiosity about users and greater attention to connecting one’s own experiences with those of others. SA displayed a non-linear pattern, dipping slightly after the diary phase and then rising after the discussions, echoing students’ accounts of becoming more aware of the distinction between their own assumptions and possible user viewpoints. In contrast, S showed more variability: a small initial decrease followed by an increase, indicating wide individual differences in immersion and sensitivity. These differing trajectories illustrate that the four dimensions did not move uniformly and that empathy-related understanding developed unevenly across the activity.

Patterns in the standard deviations further underscore these differences (Figure 1b). EI and PE both showed notable reductions in dispersion by Stage 3, indicating greater alignment in students’ self-assessments of curiosity and resonance. SA also converged substantially, while S continued to diverge, reflecting the mixed experiences students described when imagining immersion in another person’s situation. Consistent with the updated EMPA-D findings [8], which identify EI, PE, and SA as the most stable factors, our analysis showed decreasing dispersion in these three dimensions across

stages. Taken together, these descriptive patterns complement the qualitative reflections and indicate that students' empathy-related understanding shifted in multiple, non-uniform ways during the diary and discussion process.

4.3 Movement from Function-centered to Holistic Product Understanding

Lastly, we observed a distinction between how students thought about functions and how they thought about the overall product experience. Many students initially approached the application by examining whether individual features worked as expected, focusing on correctness, stability, and technical behavior. This function-oriented perspective was evident in the early diary entries, where students often checked whether the app “worked” before considering broader aspects of use. As the diary study progressed, however, students increasingly noted that functional correctness alone does not account for real user needs. A product is more than a set of working features; it also involves emotions, motivations, and values that shape how people experience technology. The FGDs further reinforced this awareness. Students noted that even when a function performs well, its meaning and impact may change when placed in a broader product context. One student emphasized the importance of aligning software and hardware:

“I used to focus on improving AI efficiency, assuming that more advanced technology would ensure better user experience. Through the diary study, I realized that users with limited hardware, such as older adults who may not buy high-end devices, face lag and long waits, which can reduce interest and create negative attitudes toward the product.”

These reflections show that students became more attentive to the tension between individual functions and the overall product experience, and that they began to consider product-level factors that influence how users actually interact with technology.

5 Discussion

5.1 Diary and Discussion as Complementary Reflective Processes

The diary and discussion activities served as complementary forms of reflection, illuminating different aspects of how students reasoned about user experience. The diary entries gave students extended opportunities to examine their own interactions with the application, capture emotional or practical challenges, and articulate moments of satisfaction or frustration. This individual perspective encouraged attention to detail and prompted students to revisit their assumptions about what users might need or value. At the same time, relying solely on personal experience risked treating one's own reactions as representative, a tendency noted widely in design education [31]. The focus group discussions provided a contrasting form of reflection by introducing diverse accounts of the same features and revealing how different backgrounds or habits lead to different interpretations. Hearing these contrasting examples helped students recognize the limits of individual experience and brought attention to scenarios or user groups they had

not previously considered. Taken together, the two activities created a setting in which students could move between examining their own experiences and comparing them with peers', offering complementary insights into user needs and UX complexity.

Beyond these immediate contrasts, the dual reflective structure appeared to create a rhythm of perspective-taking that unfolded over time. Students repeatedly moved from inward-looking reflection to outward-looking comparison, which made moments of reinterpretation more likely to surface. Several participants noted that insights that seemed straightforward at first became more nuanced when contrasted with others' experiences, broadening their understanding of how users may interpret the same feature differently. This pattern suggests that the combination of temporal distance (through diaries) and social contrast (through discussion) played a meaningful role in reshaping how students interpreted UX. The oscillation between the personal and the collective did not resolve differences but instead made them visible, providing a foundation for nuanced sensemaking about user diversity and contextual variability.

5.2 Implications for HCI Education

The paired use of diaries and peer discussions offers several implications for early HCI learning. The activities encouraged students to articulate concrete observations from everyday interactions and then revisit them in conversation with others, a process that aligns with the iterative nature of design work without replicating professional practice. This movement between individual and collective reflection also highlighted the gap between functional evaluation and broader interpretations of product experience, prompting students to consider factors such as context, hardware constraints, and emotional responses. While the study involved a small group of participants and a single application context, the descriptive patterns suggest that providing opportunities for both personal and social forms of reflection may help beginners attend to multiple perspectives when reasoning about users. Future work could explore how similar activities function in larger cohorts, across different application domains, or in courses with more diverse learner backgrounds.

6 Conclusion

This paper examined how diary reflections and peer discussions offered complementary perspectives for unpacking empathy development in early HCI learning. The combination of longitudinal self-reflection and collective comparison surfaced patterns in how students questioned their assumptions, interpreted diverse experiences, and navigated differences across the empathy dimensions. Descriptive trends from the Empathy in Design Scale, together with students' qualitative accounts, pointed to uneven and evolving forms of empathy-related understanding and highlighted the gap between initial expectations and the realities of user experience. These insights contribute to an emerging picture of how individual and group-based reflection can shape the ways learners reason about users. Future work can extend this exploration to larger cohorts and varied educational contexts to further examine the dynamics of empathy development in HCI education.

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